Indian Institute of Technology, Gandhinagar
Fluid Dynamics Laboratory (ME 207)
Type-a Experiments

Lab Experiment-I : The Pitot Tube

Objectives

(i) Study the difference between static and stagnation pressure for various flow rates.

(ii) Determine the velocity from Bernoulli's equation.

(iii) Interpret the results and the potential discrepancies between theoretical analysis and observed data.

The Essential Background

1. Read up on Bernoulli's equation in Fluid Mechanics. Fundamentally, what is Bernoulli's equation about? What are the assumptions under which this equation is valid?

2. Define Static pressure and Stagnation pressure in a flow. How can these two be used to determine the velocity?

3. What is head loss in a pipe flows (or, internal flows)? Why is it important? Can you quantify head loss?

4. What is kinetic energy coefficient in an internal flow? Can the Bernoulli's Equation be modified to account for non-uniform velocity profile and head losses due to viscosity?

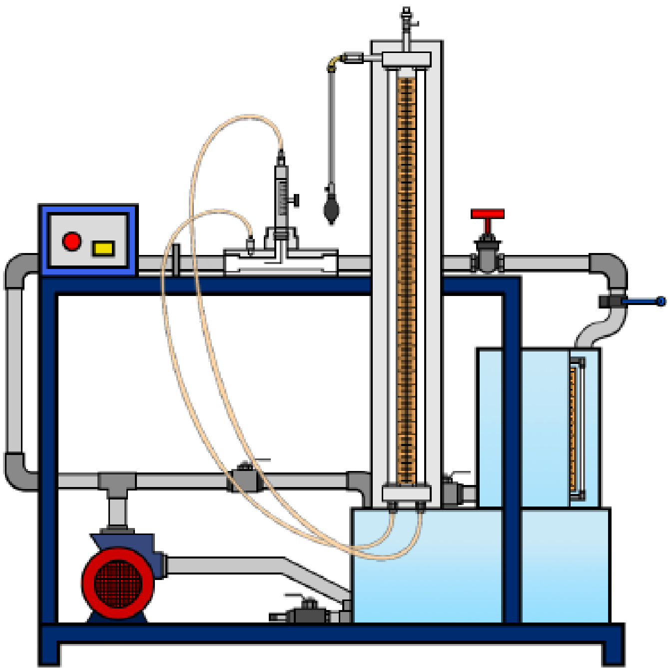

Figure 1: A schematic of the experimental set up with a pitot tube.

Task 1.1: In the report for Lab Exp.-I , concise answers to all the above questions have to be submitted. A schematic of the experimental set up has been shown on the previous page. The set up has also been explained to all the students during the first lab class. Students may take help from the lab staff and the TAs while carrying out the experiments themselves. Carry out the experiment for at least 5 different flow rates and for each flow rate take multiple readings.

Task 1.2:

(a) From the data that has been recorded by you, check whether the same satisfies Bernoulli's equation. Include the observed results in your report.

(b) If not, state what factors might cause a mismatch between theory and experimental data.

(c) Based on the "Essential Background" questions given at the beginning of this document, try to quantify the head loss from the experimental data.

(d) You have to clearly outline the procedure followed in performing the above tasks, including the numerical values of various quantities associated with the calculation.

Tasks 1.1 and 1.2 should be included in the report for this experiment.